Grapevine phenology across environments

Identify regional baselines · Improve stage-date estimates

1 Compare ten process structures

Forcing only



Heat

Sequential



Chilling + heat

Parallel



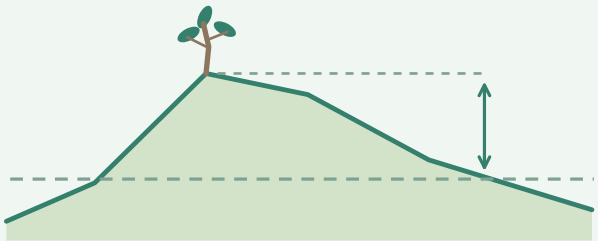
Chilling + heat

Grouped site holdouts within five environmental zones

2 Test both corrections across all ten structures

Terrain-adjusted temperature

Site–grid elevation difference



RAW ↔ Elevation-corrected

Recalibrate each process structure

Process-guided residual model

Process dates + phase duration
Weather + day length + terrain

Learn date residuals

Target: observed − process date
Out-of-fold calibration predictions

Adjust attained stage dates

3 Evaluate final accuracy and paired gains

Budburst



Flowering



Veraison



Maturity / harvest

